

Linyi (Tiger) Hou

✉ linyh2@illinois.edu | github.com/TigerHou2 | Permanent Resident

EDUCATION

Ph.D. in Aerospace Engineering, University of Illinois Urbana-Champaign Jun 2021 - May 2025
B.S. in AE, Minor in CompE, University of Illinois Urbana-Champaign Aug 2017 - May 2021

SKILLS

Technical astrodynamics, celestial navigation, modeling & simulation, data analysis
Tools/Software GMAT, FreeFlyer, Git, GitHub
Programming proficient: Python, C/C++, MATLAB | beginner: Javascript, Julia

EXPERIENCE

Graduate Researcher, University of Illinois Urbana-Champaign Jun 2021 - present

- ❖ Developed several novel spacecraft navigation and attitude determination techniques.
- ❖ Used astrodynamics toolkits in Python to simulate and analyze spacecraft navigation performance for cislunar and heliocentric missions using Monte Carlo techniques.

Teaching Assistant, University of Illinois Urbana-Champaign Jun 2021 - May 2024

- ❖ Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- ❖ Assisted in developing code and instructions for a UAV lab based on C and Python.

Flight Dynamics Related Coursework 2021 - 2024

- ❖ [AE 598, Planetary Entry] Created a 3-DOF entry simulation to assess the flight performance of modified Apollo entry guidance algorithms at Mars.
- ❖ [AE 508, Optimal Space Trajectories] Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories.

HONORS AND AWARDS

Yee Fellowship 2021, 2024
Grainger College of Engineering, University of Illinois Urbana-Champaign

Aerospace Engineering Alumni Advisory Board Fellowship 2024
Department of Aerospace Engineering, University of Illinois Urbana-Champaign

Best Student Paper, 46th AAS GN&C Conference 2024
American Astronautical Society (AAS) Rocky Mountain Section

David and Catherine Thompson Space Technology Scholarship 2020
American Institute of Aeronautics and Astronautics (AIAA)